

Brain Computer Interface Using EEG Signals and Reinforcement Learning

B.Tech. Project Report

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Abstract

Brain-Computer Interfaces (BCIs) provide a direct communication pathway between the human brain and external systems by decoding neural activity. This project focuses on the development of a non-invasive EEG-based BCI system using deep learning and reinforcement learning methodologies. EEG signals were acquired using the X.on 7-channel wireless EEG headset and processed using the EEGNet architecture for neural signal classification.

The study evaluates EEGNet on both benchmark datasets and custom datasets collected from human subjects performing directional prediction and motor-intention tasks. The project investigates the temporal characteristics of EEG signals and demonstrates that neural activity immediately after stimulus presentation contains significantly more discriminative information than signals recorded after movement execution.

Additionally, a prototype brain-controlled gaming environment was developed using reinforcement learning techniques based on Proximal Policy Optimization (PPO). The integration of EEG decoding with real-time interaction demonstrates the feasibility of EEG-based control systems for assistive technologies, gaming, and future prosthetic applications.

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Chapter 1

Introduction

Brain-Computer Interfaces (BCIs) are systems that enable communication between the human brain and external devices without relying on conventional neuromuscular pathways. Electroencephalography (EEG) is one of the most widely used non-invasive techniques for recording neural activity due to its high temporal resolution, portability, and comparatively low cost.

The objective of this project is to investigate the feasibility of decoding user intentions from EEG signals using deep learning architectures and reinforcement learning frameworks. The project explores the ability of EEGNet to capture temporal and spatial neural patterns associated with visual attention and motor preparation.

The work also investigates the effectiveness of consumer-grade EEG hardware for practical BCI applications and evaluates whether meaningful neural representations can be extracted using low-channel EEG systems.

Chapter 2

Project Objectives

The primary objectives of this project are as follows:

1. Learn EEG signal acquisition using the X.on EEG headset.
2. Study EEG signal classification using the EEGNet deep learning architecture.
3. Evaluate EEGNet on benchmark datasets including P300 and ERN paradigms.
4. Create custom EEG datasets using human subjects performing motor-intention tasks.
5. Compare EEGNet performance against traditional machine learning models such as Support Vector Machines and Logistic Regression.
6. Develop a prototype BCI-controlled game using reinforcement learning techniques.

Chapter 3

Understanding EEG Signals

Electroencephalography (EEG) records electrical activity generated by neuronal populations in the brain. EEG signals are acquired using electrodes placed on the scalp according to standardized positioning systems such as the 10-20 system.

The X.on wireless EEG headset used in this project provides seven EEG channels positioned at:

F3, F4, C3, Cz, C4, P3, and P4

EEG signals are categorized into distinct frequency bands associated with different cognitive and physiological states.

Table 3.1: EEG Frequency Bands

Band	Frequency Range	Associated Functions
Delta	0.5 – 4 Hz	Deep sleep and unconscious processes
Theta	4 – 8 Hz	Memory, meditation, learning
Alpha	8 – 13 Hz	Relaxation and calmness
Beta	13 – 30 Hz	Active thinking and concentration
Gamma	30 – 45+ Hz	High-level cognitive processing

Chapter 4

Hardware and Data Acquisition

The EEG signals were recorded using the X.on 7-channel wireless EEG headset. The headset streams EEG data through Bluetooth Low Energy and supports real-time Lab Streaming Layer (LSL) integration.

Key hardware specifications include:

- 7 EEG channels
- Sampling rates of 125 Hz, 250 Hz, and 500 Hz
- Bluetooth Low Energy transmission
- BrainVision Core Data Format support
- Real-time LSL streaming

The experimental protocol involved subjects observing directional arrow cues and attempting to predict upcoming directions. EEG recordings were synchronized with event markers using LSL for accurate temporal segmentation.

Chapter 5

EEGNet Architecture

EEGNet is a compact convolutional neural network specifically designed for EEG signal classification. It efficiently captures temporal and spatial patterns while maintaining a low parameter count suitable for small EEG datasets.

The architecture consists of the following major components:

5.1 Temporal Convolution

Temporal convolutions learn frequency-dependent features and perform band-pass filtering directly on EEG signals.

5.2 Depthwise Convolution

Depthwise convolutions learn spatial relationships across EEG electrodes while significantly reducing the number of trainable parameters.

5.3 Normalization and Regularization

Batch normalization stabilizes training while dropout and average pooling reduce overfitting and improve generalization.

Chapter 6

Benchmark Datasets

6.1 P300 Paradigm

The P300 paradigm is an oddball task in which subjects focus on target stimuli while random visual flashes are presented. A positive deflection approximately 300 ms after target presentation characterizes the P300 response.

6.2 ERN Paradigm

Error-Related Negativity (ERN) refers to a negative EEG deflection occurring shortly after an erroneous response. It is associated with activity in the anterior cingulate cortex.

6.3 X.on-EMG Human Motor Activity Dataset

A multimodal EEG-EMG dataset designed for motor intention and movement classification during bicep curl tasks. The dataset contains synchronized EEG recordings from the X.on headset and EMG signals captured during alternating curl and rest phases for brain-computer interface research and neural decoding applications.

Chapter 7

Reinforcement Learning with PPO

Proximal Policy Optimization (PPO) is a reinforcement learning algorithm designed for stable policy updates and efficient learning.

The PPO workflow consists of:

1. Collecting trajectories through environment interaction
2. Estimating the advantage function
3. Updating the policy using clipped objective functions
4. Iteratively improving agent behavior

In this project, PPO was integrated with EEG predictions to enable real-time game interaction based on decoded neural intent.

Chapter 8

Methodology

8.1 Data Acquisition

EEG signals were recorded while participants observed directional visual stimuli and attempted to anticipate subsequent directions.

8.2 Preprocessing

The acquired EEG data was segmented into temporal windows, labeled according to stimulus events, and normalized for model training.

8.3 Model Training

EEGNet models were trained on both benchmark datasets and custom datasets to classify motor intentions and directional predictions.

8.4 Evaluation

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrices

Chapter 9

Implementation

9.1 Phase 1: EEG Data Acquisition

EEG signals were collected using the X.on headset while participants observed directional cues and attempted directional prediction tasks.

9.2 Phase 2: Signal Processing and Training

The recorded EEG signals were segmented, labeled, preprocessed, and used to train EEGNet classification models.

9.3 Phase 3: Real-Time Game Integration

The trained models were integrated into a prototype EEG-controlled gaming environment for real-time interaction.

Chapter 10

Research Hypothesis

The project investigated the following hypotheses:

- Motor intention EEG signals are more informative than post-movement signals.
- EEGNet effectively captures temporal neural patterns associated with motor preparation.
- Neural information density decreases with time after stimulus presentation.
- Low-channel EEG systems can still capture meaningful motor-intention information.

Chapter 11

Results and Discussion

11.1 Performance Metrics

Table 11.1: Performance on Custom EEG Dataset

Metric	Post-Movement	Post-Stimulus
Accuracy	0.5729	0.6771
Precision	0.5591	0.6786
Recall	0.5729	0.6771
F1 Score	0.5225	0.6777

The results demonstrate that EEG signals immediately after stimulus presentation contain significantly richer information than signals acquired after movement execution.

11.2 Confusion Matrices

11.2.1 Confusion Matrix: P300 EEG

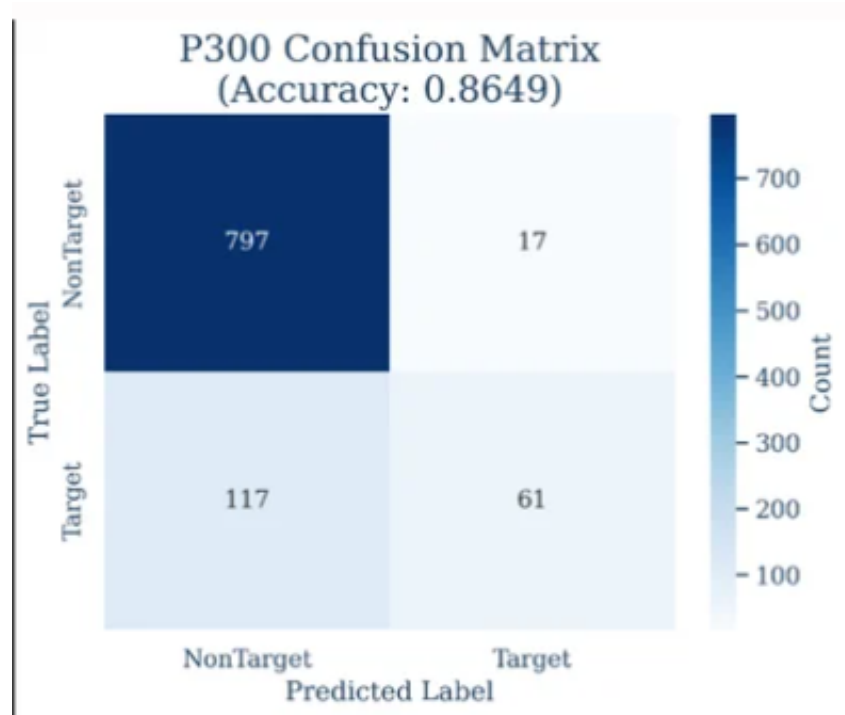


Figure 11.1: Confusion Matrix for Post-Movement EEG Classification

11.2.2 Confusion Matrix: ERN EEG

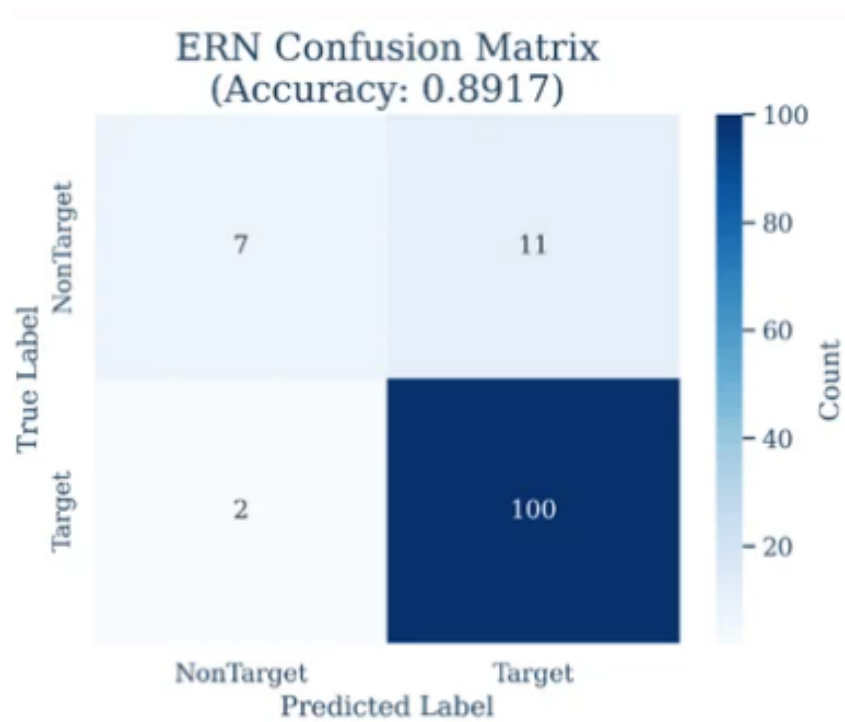


Figure 11.2: Confusion Matrix for Post-Movement EEG Classification

Chapter 12

Conclusion

This project successfully demonstrated the feasibility of decoding user intentions from EEG signals using deep learning approaches. EEGNet effectively captured temporal and spatial neural patterns associated with visual attention and motor preparation.

The study revealed that EEG signals immediately after stimulus presentation contain significantly more informative neural representations than signals recorded after movement execution. This finding is particularly relevant for real-time BCI systems and prosthetic control applications.

Furthermore, the integration of EEG decoding with reinforcement learning enabled the development of a prototype brain-controlled game, demonstrating the practical potential of EEG-driven human-computer interaction systems.

Future work may involve larger datasets, higher-density EEG systems, transformer-based architectures, and real-time adaptive prosthetic interfaces.

References

1. Lawhern, V. J., et al. “EEGNet: A Compact Convolutional Neural Network for EEG-based Brain–Computer Interfaces.”
2. Schulman, J., et al. “Proximal Policy Optimization Algorithms.”
3. X.on EEG Headset Technical Documentation.
4. Lab Streaming Layer (LSL) Documentation.